

Engineering Electromagnetics II
Homework #4 Solutions
88 points total

ECE 332 – Spring 2026

Due Friday, June 5, 2026 at 10 pm

1. A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)

At 1 MHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10^6 \text{ s}^{-1}} = 300 \text{ m}.$$

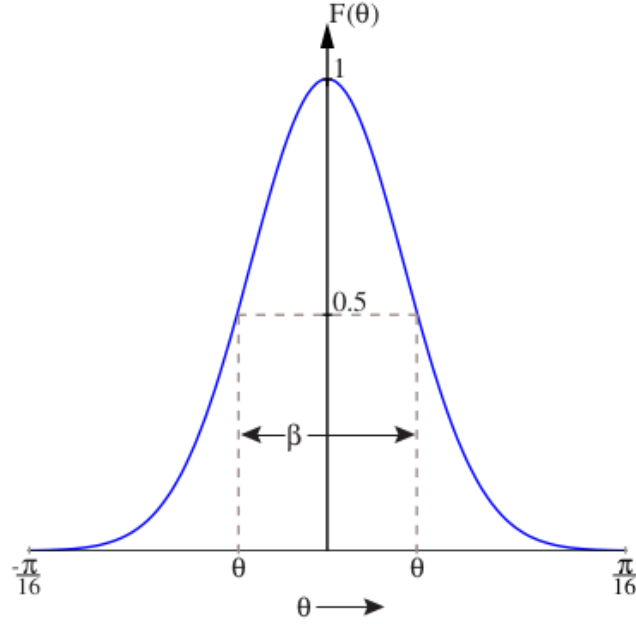
So, $l/\lambda = (1 \text{ m})/(300 \text{ m}) < 1/50$, which means that the antenna is a Hertzian dipole. Therefore,

$$\begin{aligned} S(R, \theta) &= \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta \\ &= \frac{(120\pi)(2\pi/300)^2 (12)^2 (1)^2}{32\pi^2 (5 \times 10^3)^2} \sin^2\left(\frac{\pi}{4}\right) = 1.51 \times 10^{-9} \text{ W/m}^2. \end{aligned}$$

2. The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \leq \theta \leq \pi,$$

where θ is in radians. Determine: (15 points total)



- the half power beamwidth, (5 points)

Since $F(\theta)$ is independent of ϕ , the beam is symmetrical about $z = 0$. The beam has half power when,

$$\exp(-20\theta^2) = F(\theta) = 0.5.$$

This means

$$-20\theta^2 = \ln(0.5) \implies \theta = \sqrt{-\ln(0.5)/20} = \pm 0.186 \text{ rad.}$$

So,

$$\beta = 2\theta = 0.372 \text{ rad} = 21.31^\circ.$$

- the pattern solid angle, (5 points)

The pattern solid angle is

$$\begin{aligned} \Omega_p &= \int \int_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi \\ &= 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta. \end{aligned}$$

This integral can be evaluated using software to determine

$$\Omega_p = 0.156 \text{ sr.}$$

- the antenna directivity. (5 points)

Here,

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156} = 80.55.$$

3. For a dipole antenna of length $l = \lambda/4$, (15 points total)

- determine the directions of maximum radiation, (5 points)

For an arbitrary length dipole,

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right)^2.$$

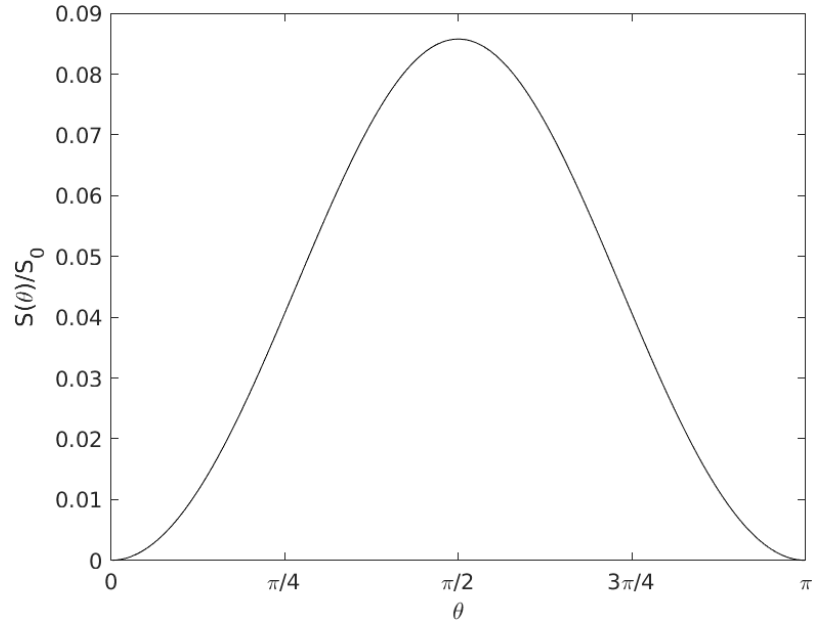
When $l = \lambda/4$, $\pi l/\lambda = \pi/4$ and

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos\left(\frac{\pi}{4}\right)}{\sin \theta} \right)^2 = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2.$$

For given values of I_0 and R , the factor $S_0 = 15I_0^2/(\pi R^2)$ is a constant and maximizing $S(\theta)$ with respect to θ means maximizing

$$\frac{S(\theta)}{S_0} = \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2$$

for $0 \leq \theta \leq \pi$. A plot of $S(\theta)/S_0$ is shown below.



The usual methods from calculus or numerical optimization show that $S(\theta)$ reaches its maximum value at $\theta = \pi/2$.

- obtain an expression for $S_{\max}$, (5 points)

The average power density reaches its maximum value $S_{\max}$ at $\theta = \pi/2$, i.e.

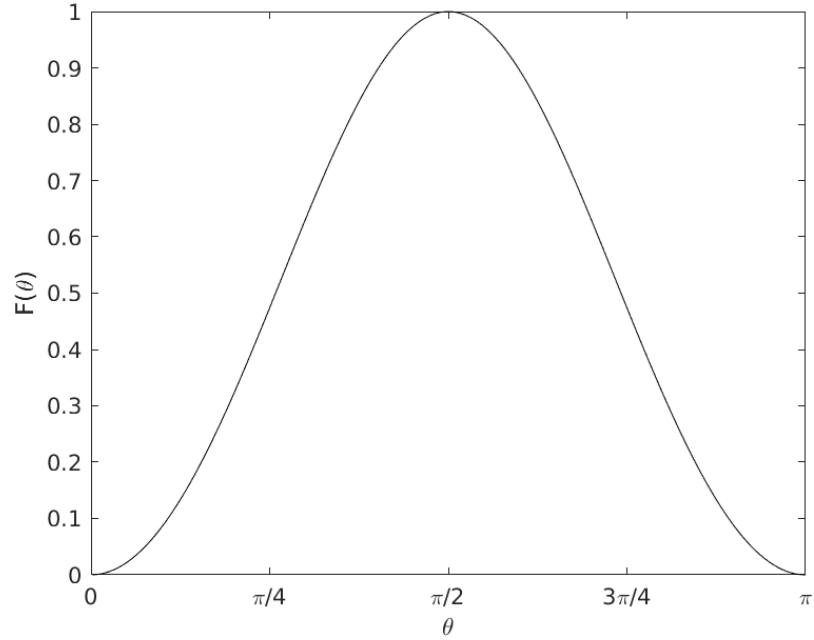
$$\begin{aligned}
 S_{\max} = S(\pi/2) &= \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{2}\right) - \frac{\sqrt{2}}{2}}{\sin\left(\frac{\pi}{2}\right)} \right)^2 \\
 &= \frac{15I_0^2}{\pi R^2} \left(\frac{\cos(0) - \frac{\sqrt{2}}{2}}{1} \right)^2 = \frac{15I_0^2}{\pi R^2} \left(1 - \frac{\sqrt{2}}{2} \right)^2 \\
 &= 0.0858 \left(\frac{15I_0^2}{\pi R^2} \right).
 \end{aligned}$$

- generate a plot of the normalized radiation pattern $F(\theta)$. (5 points)

In this case, the normalized radiation pattern is

$$F(\theta) = \frac{S(\theta)}{S_{\max}} = \frac{1}{0.0858} \left(\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right)^2,$$

which is shown below.



4. Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

In this case, $f = 100$ MHz and

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10^8 \text{ s}^{-1}} = 3 \text{ m}.$$

The directivity of a half-wave dipole antenna is $D = 1.64$. So, the effective area is

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{1.64(9 \text{ m}^2)}{4\pi} = 1.17 \text{ m}^2.$$

By comparison, the physical cross section is

$$A_p = ld = \frac{1}{2}\lambda d = \frac{1}{2}(3 \text{ m})(2 \times 10^{-2} \text{ m}) = 3 \times 10^{-2} \text{ m}^2.$$

So,

$$\frac{A_e}{A_p} = \frac{1.17}{3 \times 10^{-2}} = 39.2.$$

5. A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)

Here, $f = 5 \times 10^7 \text{ s}^{-1}$ and

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{5 \times 10^7 \text{ s}^{-1}} = 6 \text{ m}.$$

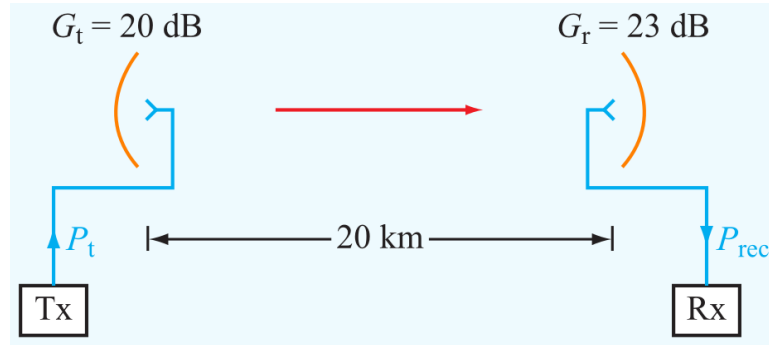
If the transmitter is a vertical half-wave dipole antenna, then its gain is $G_t = 1.64$. The gain of the receiving antenna is

$$G_r = 3 \text{ dB} = 10^{3/10} = 2.$$

Assuming that the home antenna and the transmitting antenna are properly aligned,

$$\begin{aligned} P_{\text{rec}} &= G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t = (1.64)(2) \left(\frac{6 \text{ m}}{4\pi(3 \times 10^4 \text{ m})} \right)^2 (10^3 \text{ W}) \\ &= 8.3 \times 10^{-7} \text{ W}. \end{aligned}$$

6. Consider the communication system with all components properly matched shown below.



Determine the following if $P_t = 10 \text{ W}$ and $f = 6 \text{ GHz}$. (15 points total)

- (a) What is the power density at the receiving antenna (assuming proper alignment of antennas)? (5 points)

When the antennas are properly aligned, the power density at the receiving antenna is

$$S_r = G_t S_{\text{iso}} = G_t \frac{P_t}{4\pi R^2}.$$

Here,

$$G_t = 20 \text{ dB} \implies G_t = 10^{20/10} = 10^2 = 100.$$

So,

$$S_r = 100 \frac{10 \text{ W}}{4\pi (20 \times 10^3 \text{ m})^2} = \frac{1}{8\pi} \times 10^{-5} \text{ W/m}^2 \approx 3.98 \times 10^{-7} \text{ W/m}^2.$$

- (b) What is the received power? (5 points)

The received power is

$$P_{\text{rec}} = \xi_r P_{\text{int}} = \xi_r S_r A_r = \xi_r S_r \frac{\lambda^2 G_r}{4\pi \xi_r} = \frac{\lambda^2 G_r}{4\pi} S_r$$

In this case,

$$G_r = 23 \text{ dB} \implies G_r = 10^{23/10} = 10^{2.3} \approx 199.526$$

and, assuming that the medium is air,

$$\lambda = \frac{u_p}{f} = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{6 \times 10^8 \text{ s}^{-1}} = \frac{1}{2} \text{ m}.$$

Altogether, the received power is

$$P_{\text{rec}} = \frac{(0.25 \text{ m})^2 199.526}{4\pi} \left(\frac{1}{8\pi} \times 10^{-5} \text{ W/m}^2 \right) \approx 3.95 \times 10^{-7} \text{ W}.$$

- (c) If $T_{\text{sys}} = 1000 \text{ K}$ and the receiver bandwidth is 20 MHz , what is the signal-to-noise ratio in decibels? (5 points)

The noise power is

$$\begin{aligned} P_n &= KT_{\text{sys}}B = (1.38 \times 10^{-23} \text{ J/K})(1000 \text{ K})(20 \times 10^6 \text{ s}^{-1}) \\ &= 2.76 \times 10^{-13} \text{ W}. \end{aligned}$$

Therefore, the signal-to-noise ratio is

$$\frac{P_{\text{rec}}}{P_n} = \frac{3.95 \times 10^{-7} \text{ W}}{2.76 \times 10^{-13} \text{ W}} = 1.43 \times 10^6.$$

Expressed in dB, this is

$$\frac{P_{\text{rec}}}{P_n} \text{ dB} = 10 \log(1.43 \times 10^6) \text{ dB} \approx 61.6 \text{ dB}.$$

7. An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz. (15 points total)

- (a) What is the antenna directivity in dB? (5 points)

A circular aperture has equal bandwidths in the x - z and y - z planes, i.e.

$$\beta_{xz} = \beta_{yz} = \beta = 3^\circ = 3^\circ \frac{\pi \text{ rad}}{180^\circ} = \frac{\pi}{60} \text{ rad at 20 GHz}.$$

The directivity is

$$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi}{(\pi/60)^2} = \frac{14400}{\pi} \approx 4584.$$

Expressed in dB, this is

$$D \approx 10 \log(4584) \text{ dB} \approx 36.6 \text{ dB}.$$

- (b) If the antenna area is doubled, what will be the new directivity and new beamwidth? (5 points)

Since

$$D \approx \frac{4\pi A_p}{\lambda^2},$$

doubling the antenna area will double the directivity. This means that the new directivity will be

$$D \approx 2(4584) = 9169 \implies D \text{ dB} \approx 10 \log(9169) \text{ dB} \approx 39.6 \text{ dB}.$$

Also,

$$D \approx \frac{4\pi}{\beta^2} \implies \beta^2 \approx \frac{4\pi}{D} \implies \beta \approx 2\sqrt{\frac{\pi}{D}}.$$

So, doubling the directivity will divide the bandwidth by $\sqrt{2}$. In this case, the new bandwidth will be

$$\beta = \frac{3^\circ}{\sqrt{2}} \approx 2.12^\circ \approx 0.0370 \text{ rad}.$$

- (c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then? (5 points)

Since

$$\lambda = \frac{u_p}{f},$$

doubling the frequency will divide the wavelength λ by 2. Also,

$$D \approx \frac{4\pi A_p}{\lambda^2}.$$

So, doubling the frequency will divide the directivity by 4, i.e. the new directivity will be

$$D \approx \frac{1}{4}(4584) \approx 1146 \implies D \text{ dB} \approx 10 \log(1146) \approx 30.6 \text{ dB}.$$

In addition,

$$\beta \approx 2\sqrt{\frac{\pi}{D}}$$

means that dividing the directivity by 4 will multiply the beamwidth by 2 so that the new beamwidth will be

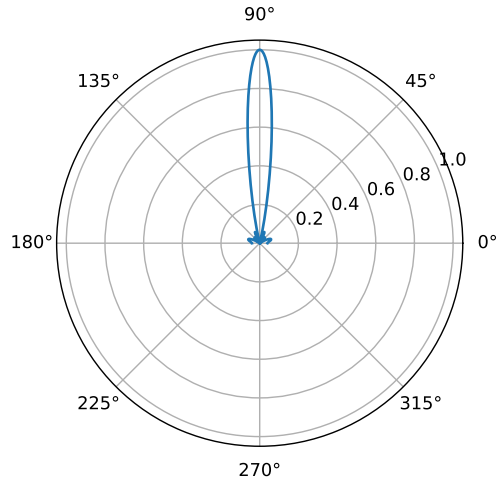
$$\beta = 6^\circ = 6^\circ \frac{\pi \text{ rad}}{180^\circ} = \frac{\pi}{30} \text{ rad} \approx 0.105 \text{ rad}.$$

8. Find and plot the normalized array factor and determine the half-power beamwidth for a 5-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is $3\lambda/4$. (10 points)

For a 5-element linear array excited with equal phase and a uniform amplitude distribution with $d = 3\lambda/4$, the normalized array factor is

$$F_{an} = \frac{\sin^2\left(\frac{5\pi(3\lambda/4)}{\lambda} \cos \theta\right)}{25 \sin^2\left(\frac{\pi(3\lambda/4)}{\lambda} \cos \theta\right)} = \frac{\sin^2\left(\frac{15\pi}{4} \cos \theta\right)}{25 \sin^2\left(\frac{3\pi}{4} \cos \theta\right)}.$$

A plot of F_{an} for $0 \leq \theta \leq \pi$ is shown below.



Code for creating this plot is in Appendix A.

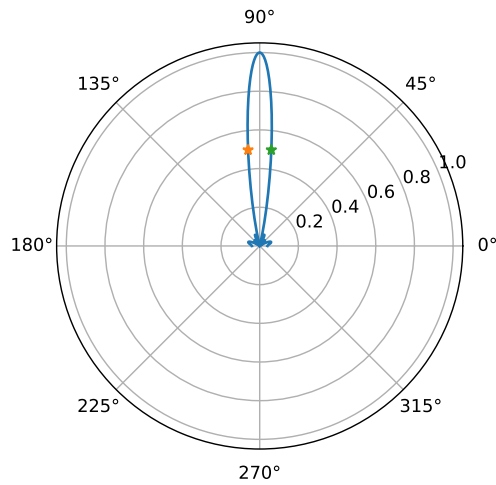
Assuming that the power density of an individual antenna doesn't depend on θ , the power density is half of its maximum power when F_{an} is half its maximum value. This occurs at the angles $\theta = \frac{\pi}{2} \pm \Delta\theta$ satisfying

$$\frac{\sin^2\left(\frac{15\pi}{4} \cos\left(\frac{\pi}{2} \pm \Delta\theta\right)\right)}{25 \sin^2\left(\frac{3\pi}{4} \cos\left(\frac{\pi}{2} \pm \Delta\theta\right)\right)} = \frac{1}{2} \quad (1)$$

Code for using the SciPy function `fsolve` to find $\Delta\theta$ is in Appendix A. The half-power beamwidth is

$$\beta = 2\Delta\theta \approx 0.241 \text{ rad} \approx 13.8^\circ$$

and the angles on the main lobe at the points where the power density is half its maximum are shown in the figure below.



A Python Code for Problem 8

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve

theta = np.linspace(0.0, np.pi, 10000)
threePiOverFour = 0.75*np.pi
sinArg = threePiOverFour*np.cos(theta)
numSin = np.sin(5.0*sinArg)
denomSin = np.sin(sinArg)
ratio = np.divide(numSin, denomSin)
Fan = (1.0/25.0)*np.multiply(ratio, ratio)

plt.polar(theta, Fan)
plt.grid(True)
plt.tight_layout()
plt.savefig('prob8.pdf')
plt.show()

def f(dtheta):
    piOverTwo = 0.5*np.pi
    threePiOverFour = 0.75*np.pi
```

```

    arg = threePiOverFour*np.cos(piOverTwo + dtheta)
    numTerm = np.sin(5.0*arg)
    denomTerm = np.sin(arg)
    r = numTerm/denomTerm
    return (1.0/25.0)*r*r - 0.5

deltaAngle = fsolve(f, 0.00001)
print(2.0*deltaAngle) # This is the half-power beamwidth.
print(f(deltaAngle))

angleUpper = 0.5*np.pi + deltaAngle
angleLower = 0.5*np.pi - deltaAngle

plt.polar(theta, Fan, angleUpper, 0.5, '*', angleLower, 0.5, '*')
plt.grid(True)
plt.tight_layout()
plt.savefig('prob8halfPower.pdf')
plt.show()

```